LOVON: Legged Open Vocabulary Object Navigator

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***Abstract*—Object navigation in open-world environments remains a formidable challenge for legged robots, as their inherent mechanical constraints and controller characteristics demand tight alignment between high-level navigation algorithms and low-level motion execution, particularly when it comes to executing long-horizon tasks that require both open-world object detection and high-level task planning. Traditional methods often struggle to integrate these components effectively, and this limits their capability to deal with complex, long-range navigation missions. We propose LOVON, a framework for open-vocabulary object navigation that integrates LLMs, open-vocabulary visual detection, and a language-to-motion model (L2MM) tailored to the mechanical and control characteristics of legged robots for effective long-range object navigation in dynamic, unstructured environments. To address key challenges—including motion blur induced by robot locomotion and misalignment between navigation goals and mechanical execution—we design a Laplacian Variance Filter and execution logic that guarantees LOVON's capabilities in autonomous navigation, task adaptation, and robust task completion. Extensive evaluations demonstrate the successful completion of long-sequence tasks involving real-time detection, search, and navigation toward open-vocabulary dynamic targets. Furthermore, real-world experiments across different legged robots (Unitree Go2, B2, and H1-2) showcase the compatibility and appealing plug-and-play feature of LOVON. Our code is available at https://github.com/DaojiePENG/LOVON.**

***Index Terms*—Legged Robots, Open-Vocabulary Navigation, Robotic Autonomy**

I. Introduction

In recent years, large language models (LLMs) [1] and vision models [2]–[5] have achieved revolutionary breakthroughs in the field of artificial intelligence. LLMs have significantly improved their capabilities in understanding and planning long-horizon tasks, enabling them to deeply comprehend complex contexts and generate efficient execution strategies, which brings new possibilities to task planning. Meanwhile, advances in open-vocabulary visual detection have empowered vision models to recognize and understand a diverse range of objects beyond predefined categories, greatly enhancing the adaptability of machine vision systems in scene understanding and object recognition. These leaps in perception and cognition lay a solid foundation for addressing complex long-horizon tasks in robotics.

Legged robots have evolved over decades and exhibit outstanding mobility in complex terrains, enabled by their specialized mechanical designs. However, most current research focuses on optimizing single tasks (e.g., walking, jumping, climbing) or low-level mechanical control, and short-range navigation, lacking a comprehensive consideration of complex long-horizon missions. Our work addresses this by integrating advanced language and vision understanding, enabling language-specific velocity control and long-horizon task completion in open environments.

In this paper, we propose LOVON, an innovative operating system that integrates the task planning capabilities of LLMs, the perception abilities of open-vocabulary visual detection, and a language-to-motion model (L2MM) for precise motion prediction, as shown in Fig. 1. The LOVON system addresses real-world challenges such as visual jitter caused by robot motion. In particular, we design a Laplacian Variance Filtering technique to effectively mitigate visual instability during robot movement, ensuring the accuracy and continuity of object detection. We also propose the functional execution logic for robust task completion. Experiments on simulation benchmark Gym-Unreal [6] demonstrate the superior results of LOVON. In addition, we conduct extensive experiments on multiple legged platforms (Unitree Go2, B2, and H1-2), successfully accomplishing long-horizon tasks involving real-time detection, search, and navigation toward open-vocabulary dynamic targets. To the best of our knowledge, LOVON is the first operational system to achieve such complex capabilities in unstructured environments. The main contributions of this work are summarized as follows:

- We propose LOVON, a unified framework that integrates LLMs, open-vocabulary visual detection, and a language-to-motion model, enabling the planning and execution of complex open-world long-horizon navigation tasks.
- We develop a Laplacian Variance Filtering method to resolve motion-induced blurring issues and improve the robustness of the system. Additionally, we introduce a robot execution logic that ensures adaptability to various environments.
- We conduct comprehensive validation through simulations and real-world experiments across a variety of legged robot platforms. The results demonstrate that our system successfully performs open-vocabulary object search and navigation tasks in unstructured scenes.

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Fig. 1. **Object navigation of legged robots in diverse open-world scenarios. These scenarios include long-horizon multi-subgoal sequential navigation, dynamic tracking of moving humans and objects, cross-platform deployment on different legged robots (Unitree Go2, B2, H1-2), automatic target re-localization after temporary occlusion or loss, flexible target switching for multi-object tasks, and open-vocabulary detection for arbitrary daily objects. Quantitative results further validate robust performance under varying speeds, illuminations, complex backgrounds, and mechanical disturbances such as sand jolting, demonstrating LOVON's strong practicality, stability, and generalization in real-world unstructured environments.**

II. Related Works

A. Large Language Models for Robotic Task Planning

LLMs, such as GPT [1], PaLM [7], [8], and LLaMA [9], have demonstrated remarkable capabilities in natural language understanding, reasoning, and task decomposition. In the context of robotics, LLMs have been increasingly adopted for high-level task planning, instruction following, and semantic reasoning. Recent works leverage LLMs to interpret human instructions, generate action sequences, and adapt plans to dynamic environments, enabling robots to perform complex, long-horizon tasks with greater autonomy and flexibility. For example, SayCan [10] integrates LLMs with robotic affordance models to map language instructions to executable actions, while Code as Policies [11] utilizes LLMs to generate code for robot controllers directly from natural language descriptions. Despite these advances, challenges remain in grounding language to real-world robot actions [12], [13], handling ambiguous or user-specified instructions, and ensuring robust performance in unstructured environments. Our work builds upon these foundations by integrating LLM-based planning with open-vocabulary perception and legged robot mobility, aiming to address the limitations of previous approaches in long-horizon, open-world scenarios.

B. Open-Vocabulary Visual Perception

Open-vocabulary visual perception has evolved significantly from early fixed-class object detectors like YOLO [3], which are confined to recognizing predefined categories and struggle in open-world scenarios. Modern advancements have shifted toward models that leverage cross-modal representations, such as CLIP [14], which learns to align visual and language embeddings through contrastive pretraining on massive image-text pairs, enabling zero-shot object recognition by mapping visual features to arbitrary language descriptions. Subsequent works like Grounding DINO [15] and YOLO-World [16] have further enhanced the detection capability by integrating grounded pretraining to improve open-set detection accuracy. For robotic applications, real-time performance and robustness to dynamic camera motions, such as jittering caused by legged robot locomotion or temporary target occlusions, remain critical challenges [17]. Existing methods often fail to maintain detection stability in such scenarios, lacking the feedback mechanisms to adapt to the robot's motion state or environmental changes. LOVON addresses these gaps by developing specialized preprocessing techniques to mitigate motion blur and ensure consistent visual input, while tightly integrating open-vocabulary detection with task planning and motion control for end-to-end execution in unstructured environments.

C. Legged Robot Navigation and Long-Horizon Autonomy

Legged robot navigation has advanced from low-level mobility to complex tasks, but existing approaches often focus on single-task optimization [18]–[21] and lack integration of high-level planning for long-horizon missions [22]–[24]. While legged robots show superior terrain adaptability [25], their potential for executing sequential goals in unstructured

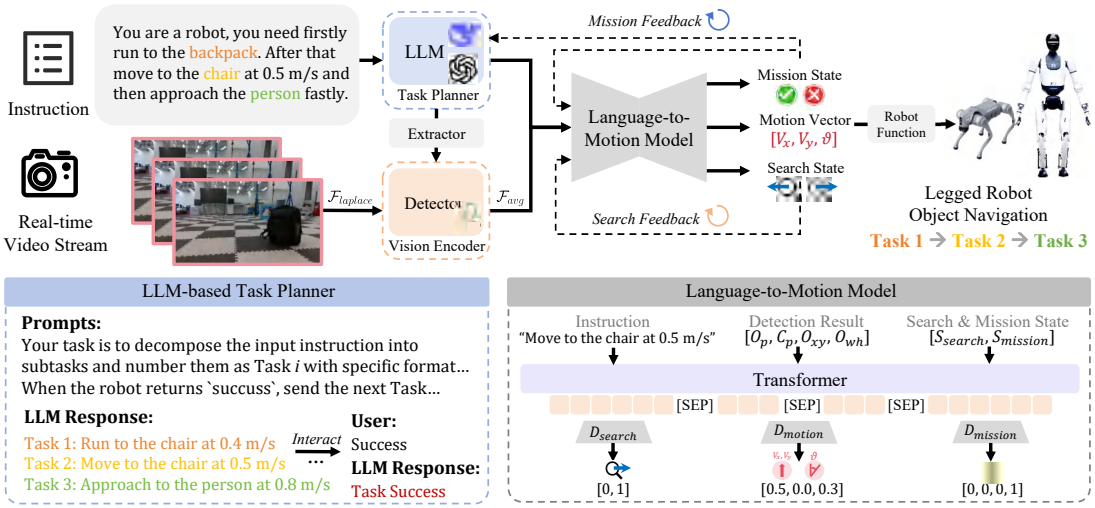


Fig. 2. **Overview of LOVON's Workflow.** The workflow operates in three core stages: **1)** Task Reconfiguration: The LLM Task Planner transforms human-defined long-horizon tasks into structured basic mission instructions. **2)** Video Stream & Instruction Object Processing: The detection model processes the video stream, integrating a Laplacian filter and instruction objects extracted via IOE. **3)** Motion Generation & Feedback: Mission instructions, target objects, bounding boxes, and robot states are input to the Language-to-Motion Model (L2MM). The L2MM generates the robot's control vector and provides feedback to both the LLM and itself, facilitating the progressive completion of all tasks.

environments remains underexplored, as traditional systems often separate perception from motion planning and struggle with dynamic target tracking. LOVON addresses this by unifying hierarchical task decomposition with real-time motion control. An LLM-based planner breaks down long-horizon tasks, while the L2MM maps instructions and visual feedback to dynamic motion vectors. This enables adaptive behavior and cross-platform validation in diverse terrains. By integrating open-vocabulary perception with legged mobility, LOVON enables autonomous robots to navigate complex and dynamic environments.

III. Problem Formulation

Task. The task involves operating in an arbitrary open-world environment, where the robot is required to perform long-horizon tasks to search for different targets. The long-horizon task T_l is defined as a set of subtasks $T_l = \{T_1, T_2, \ldots\}$, where each subtask T_i corresponds to searching for a specific target O_i. The description of T_l is flexible (example in the bottom left of Fig. 2), allowing for varying mission objectives. The core challenge is for the robot to autonomously search for and identify different subgoals (targets), navigating toward them at different velocities based on mission instructions. These subgoals can vary throughout the task, requiring the robot to adapt dynamically.

Goal. Our goal is to develop a dual-system model: (I) A high-level policy that can decompose complex task instructions T_l into individual subtasks with concrete instructions $I_{\text{ins}} = \{I_i | I_1, I_2, ...\}$ and perform task planning; (II) A low-level policy that, based on specific subtask instructions I_i and video stream input I_{RGB}, can generate motion vectors $V_m \in R^3$ to achieve precise motion control. The model should be adaptable across various legged robots, ensuring versatility in real-world applications.

IV. Methodology

A. Overview

The workflow of LOVON is illustrated in Fig. 2. Initially, the LLM reconfigures the human's long-horizon task into basic mission instructions. This decomposition ensures that high-level objectives (e.g., 'You are a robot, you need to first run to the backpack. After that move to the chair at 0.5 m/s and then approach the person quickly.') are translated into actionable, step-by-step directives suitable for subsequent robotic processing. These directive mission instructions are then transmitted to an instruction object extractor (IOE) to accurately identify and extract key target objects from the instructions. The detection model processes the real-time captured video stream, with the input image preprocessed leveraging a Laplacian filter which specifically addresses motion blur in the captured frames and the information extracted from IOE. Finally, the mission instruction, target object, bounding box, mission state, and search state are combined as inputs to the proposed L2MM, which generates the robot's control vector and feedback states for both the LLM and L2MM to enable dynamic task adjustments.

B. Multimodal Input Processing

LOVON integrates two classes of pretrained models: object detection models (e.g., [5], [15], [16], [26], [27]) for visual input processing and LLMs (e.g., [1], [28], [29]) for long-horizon task management. The input to the LLM consists of the system description I_{sys}, the user's long-sequence task description T_l, and feedback from the L2MM O_f. Using this input, the LLM generates specific mission instructions I_i, enabling LOVON to execute long-sequence tasks by producing the necessary instructions to achieve the mission objectives:

$$I_{\text{ins}} = f_{\text{LLM}}(I_{sys}, T_l, O_f). \tag{1}$$

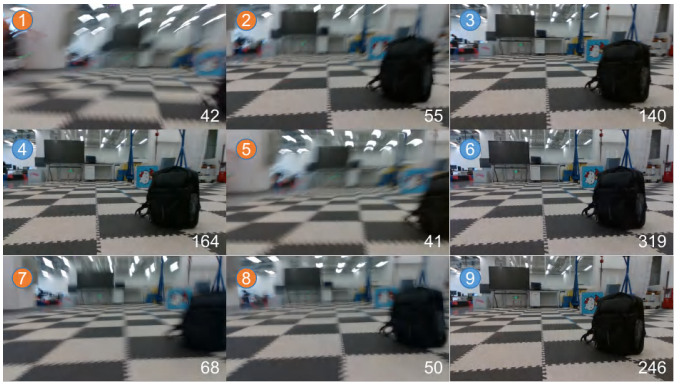


Fig. 3. **Image blurring phenomenon.** This figure shows the occurrence of image blurring in the robot view, which impacts the clarity and accuracy of the processed images. The number in the lower right corner of each image represents its Laplacian variance.

Then, our proposed IOE maps the instruction to the detection class. During deployment, L2MM operates at a high update rate, while LLM-based long-horizon task planning is executed only once. Although LLMs can handle both task planning and instruction-to-class mapping, using them for the latter is inefficient, wasting tokens, increasing inference cost, and extending latency. IOE efficiently maps instructions to detection classes, simplifying LLM prompt structure, reducing inference frequency, and enabling a more efficient, cost-effective practical solution. IOE uses a two-layer transformer with a linear layer as head to predict the object class:

$$I_{\text{object}} = f_{\text{IOE}}(I_m) \in \mathbf{C}, \tag{2}$$

where $\mathbf{C}$ represents the set of classes that the detection model is capable of recognizing.

Regarding the visual processing, the object detection model takes an RGB image I_{RGB} and I_{object} as input and outputs the desired detection information as follows:

$$O_m, C_p, O_{xy}, O_{wh} = f_{\text{det}}(I_{\text{RGB}}, I_{\text{object}}). \tag{3}$$

We use the normalized format for the detection results, with the predicted object denoted as O_m, the confidence score as C_p, and the center position of the bounding box as $O_{xy} = [x_n, y_n]$. The width and height of the bounding box are represented as $O_{wh} = [w_n, h_n]$. At the interface level, the detector and LLM planner are replaceable, provided that they return the structured detection tuple and ordered mission instructions, respectively; L2MM does not consume backbone-specific features. The implemented window-3 MAF smooths the detection confidence C_p, thereby indirectly stabilizing the bounding-box signal jointly processed by L2MM.

C. Adaptive Laplacian Variance-Based Motion Blur Filtering

The locomotion of legged robots involves inherent mechanical fluctuations (e.g., step transitions, joint vibrations, terrain-induced body sway) that propagate to onboard cameras as shown in Fig. 3. Several frames (e.g., 1/2/5/7/8) are particularly blurred due to the robot's dynamic locomotion (e.g., step transitions, balance adjustments, or movement over rough terrain). For the downstream vision model, this motion-induced visual degradation affects its ability to extract visual features, locate objects, and align visual-language information, which are critical to the performance of the vision-language workflow. To address this, we design a Laplacian Variance Filter tailored to the motion frequency of legged robots: the threshold is calibrated based on the typical mechanical vibration range of Unitree Go2/B2/H1-2, ensuring it filters blur from mechanical motion while retaining clear frames. This preprocessing guarantees high-fidelity visual inputs during dynamic locomotion, and improves the reliability of the legged robot's vision-guided system.

In particular, we first convert the RGB frame I_{RGB} to a grayscale frame I_{gray}. Next, we apply the Laplacian operator to enhance high-frequency components, generating the Laplacian response. The variance of the Laplacian response, $v_t = \text{Var}(\nabla^2 I_{\text{gray},t})$, is computed to assess the clarity of frame t. If v_t is below the current threshold $T_{\text{blur}}^{(k)}$, the frame is classified as blurred and replaced by the last clear frame. For instance, when $T_{\text{blur}} = 100$, the motion-blurred frames labeled in orange (Fig. 3) are filtered out.

The threshold is initialized with $T_{\text{blur}}^{(0)} = 150$, which is empirically calibrated from the observed results across different objects, speeds, illuminations, and backgrounds, and is evaluated on different robot platforms. To avoid treating this observation-based value as a universal constant, we adapt it using the unfiltered Laplacian variances in a preceding $\Delta t = 2$ s window (approximately 30 frames at 15 Hz). At the end of window k, the threshold is updated as

$$T_{\text{blur}}^{(k+1)} = \text{clip}\Big((1-\alpha)T_{\text{blur}}^{(k)} + \alpha q_{0.2}^{(k)},\ 0.5T_{\text{blur}}^{(0)},\ 1.5T_{\text{blur}}^{(0)}\Big), \tag{4}$$

where $q_{0.2}^{(k)}$ is the 20th percentile of the buffered variances and $\alpha = 0.2$. The lower quantile tracks the local low-sharpness tail, whereas exponential smoothing and clipping prevent abrupt changes and threshold drift. This rule adapts the rejection boundary to robot- and motion-dependent image statistics without additional training or sensors. The performance of this filtering method is discussed in Sec. VII-B.

D. Language-to-Motion Model

Our proposed L2MM bridges open-vocabulary navigation goals with legged robots' mechanical execution capabilities, enabling generalization beyond the limitations of hard-coded instructions and robust performance on real-world unstructured inputs. Unlike generic motion generators, it is trained to generate velocity control instructions ($V_m = [v_x, v_y, \theta]$) natively compatible with legged robots' built-in controllers. This design enables language-based speed control and plug-and-play deployment without hardware/controller modifications.

L2MM uses an encoder-decoder architecture designed to translate mixed inputs into robot motion commands. The encoder processes a structured sequence of diverse inputs that collectively capture task context, environmental object perception, and robot state: these include the previous mission instruction I_{m0} for new mission awareness, the current mission instruction I_{m1} for immediate objectives, the predicted objects O_p, the predicted confidences C_p, the center positions O_{xy}, the width and height of the normalized bounding boxes O_{wh},

as well as the current mission state S_m and the current search state S_s, which track task progress and object-finding status. To unify these heterogeneous inputs, each component is converted to a string format and concatenated into a single sequence using [SEP] tokens as delimiters, forming the encoder input $I_{encoder} = \{I_{m0}, I_{m1}, O_p, C_p, O_{xy}, O_{wh}, S_m, S_s\}$. This sequence is processed by the encoder, which extracts contextual relationships between inputs and outputs a compact latent state l_e. This latent state is then passed to the decoder, which features three distinct, parallel heads—each specialized to handle a separate prediction task critical for generating coherent robot motion:

Motion Vector Head D_{motion}. This head predicts the robot's motion vector V_m based on the latent state l_e. It is formulated as a sequence-to-vector problem, where the output is the control vector for the robot's movement:

$$V_m = D_{motion}(l_e). \tag{5}$$

Mission State Head $D_{mission}$. This head predicts the mission state S_m that is used as feedback for the encoder to adjust actions. The prediction is formulated as a sequence-to-number problem, where the output represents the current status of the mission. The prediction is given by:

$$S_m = D_{mission}(l_e). \tag{6}$$

Search State Head D_{search}. This head predicts the search state S_s, which indicates the robot's progress in searching for the target. It is also a sequence-to-number problem, where the output reflects the current state of the search:

$$S_s = D_{search}(l_e). \tag{7}$$

Each of these decoder heads uses a linear layer to process the latent state l_e and generate the respective outputs.

The final output of the model is a combination of the predictions from all three decoder heads:

$$O_{decoders} = \{V_m, S_m, S_s\}, \tag{8}$$

where $V_m = [v_x, v_y, \theta]$, v_x and v_y represent the robot's velocities along the x and y axes, and θ represents the angular velocity; S_m and S_s are the output mission state and search state, respectively.

This architecture enables the model to predict motion vectors, task states, and search states simultaneously, allowing the robot to not only control its motion accurately but also understand long task sequences and provide relevant feedback. Moreover, the L2MM outputs velocity vectors $V_m = [v_x, v_y, \theta]$ strictly constrained within the robot's mechatronic system executable range, ensuring the instructions are actionable; meanwhile, the L2MM's extensive navigation data training enables implicit correction of execution errors, forming a data-driven stability guarantee that maintains reliable interaction between the LLM and low-level mechatronic controllers.

E. Loss Functions

The model is trained using different loss functions depending on the task:

TABLE I

RELATIONSHIP BETWEEN STATES AND MOTION VECTOR. IN THE RUNNING STATE, v_x REPRESENTS THE MOVING SPEED OF x AXIS AND v_y REPRESENTS THE MOVING SPEED OF y AXIS. THE ROTATION θ_{corr} IS REGULATED BASED ON THE CENTER POSITION O_{xy} DURING EXECUTION TO CORRECT THE HEADING DIRECTION.

State S_m/S_s	Motion vector V_m
success	$[0, \quad 0, \quad 0]$
running	$[v_x, \ v_y, \quad \theta_{corr}]$
searching_0	$[0, \quad 0, \quad -0.3]$
searching_1	$[0, \quad 0, \quad 0.3]$

Motion Vector Loss. For the motion vector head D_{motion}, we use the mean squared error loss with coefficient β to measure the difference between the predicted and actual motion vectors:

$$L_{MSE} = \frac{1}{N}\beta \sum_{i=1}^{N} (V^i_{m_{pred}} - V^i_{m_{true}})^2. \tag{9}$$

Mission and Search State Loss. For the mission and search state heads $D_{mission}$ and D_{search}, we use cross-entropy loss to compare the predicted states with the ground truth labels:

$$L_{CE} = -\sum_{i=1}^{N} y_i \log(p_i), \tag{10}$$

where y_i is the true label, and p_i is the predicted probability for each class.

V. ROBOT FUNCTIONAL LOGIC FOR TASK EXECUTION

After model prediction, the robot executes tasks and adapts to environmental changes via predefined functional logic. The core behaviors are defined as follows:

- **Execute New Mission**: The robot initiates a new task upon detecting an instruction mismatch between the current and previous commands.
- **Run to the Object**: The robot navigates toward the detected target using the predicted motion vector and detection outputs.
- **Search for Lost Object**: If the target is lost, the robot automatically enters a searching state and adjusts its motion to re-localize the object.
- **Maintain the State**: The robot retains its current operational state based on real-time visual feedback until a trigger condition for state transition is met.
- **Accomplish the Mission**: The robot terminates movement and enters the `success` state once the target O_{wh} satisfies the predefined success threshold.

These rules govern the robot's operational state for data labeling, ensuring robust autonomous navigation and reliable task completion. As illustrated in Fig. 5, LOVON implements a finite state machine consisting of four states: `success` (task completed), `running` (approaching the target), `searching_0` (left rotation), and `searching_1` (right rotation). State transitions are triggered by task inconsistency (`task_0`$\neq$`task_1`), detection confidence thresholds, and the previous search state. Specifically, the system switches between the two search modes when confidence drops below 0.25, enters `running` when confidence is sufficient, and

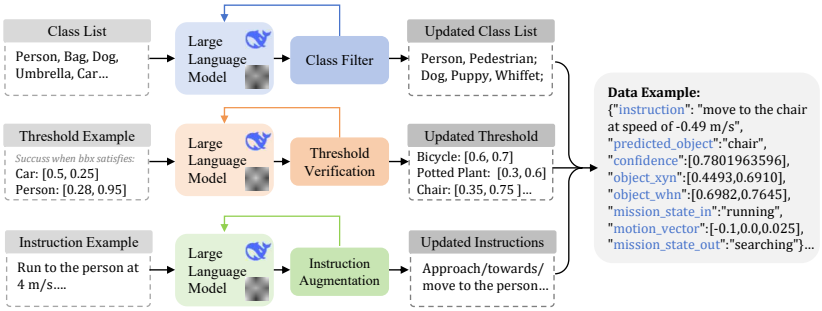


Fig. 4. **Illustration of the dataset preparation pipeline**, including four key steps and an iterative optimization loop. **1)** Detection Class Synonym Expansion: LLM generates object class synonyms to enrich categories and generalization; **2)** Instruction Variation: LLM produces paraphrases with diverse structures while preserving core semantics; **3)** Threshold Generation: Defines initial detection thresholds and uses LLM to adapt them for different object sizes; **4)** Ground-Truth Labeling: Labels robot states and motion command V_m per template constraints. An LLM-driven feedback loop iteratively refines templates, reducing redundancy and enhancing diversity.

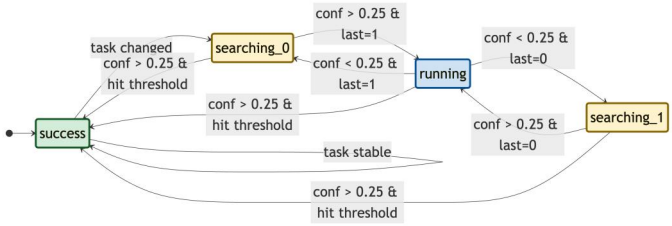


Fig. 5. **State Transition Diagram of LOVON**, including four states: success, running, searching_0, and searching_1. Transitions are triggered by task changes, detection confidence thresholds, and historical search states. The system maintains the success state until a new task discrepancy occurs, enabling robust switching between searching and running behaviors.

achieves `success` upon valid target detection. The corresponding motion vectors for each state are detailed in Table I.

VI. Dataset Preparation

As illustrated in Fig. 4, the dataset preparation pipeline constructs diverse training samples and assigns their ground-truth labels as follows:

Dataset Diversity. Diversity is introduced through four complementary sampling dimensions within each task scenario: **1) object diversity**, where an LLM generates synonymous descriptions for predefined object classes and different target classes are sampled; **2) prompt diversity**, where the LLM paraphrases mission instructions to vary their sentence structures while preserving task semantics; **3) camera-view diversity**, where the normalized bounding-box center and size are varied to represent different target directions and apparent scales in the camera view; and **4) motion diversity**, where different commanded speeds and corresponding navigation states are sampled. These dimensions are combined during sample generation so that L2MM observes varied language, perception, and motion conditions rather than a single scene configuration.

Threshold Generation for Object Categories. We define success thresholds for object detection based on initial examples, then use the LLM to adapt these thresholds for other categories, ensuring the model handles different object sizes. During dataset preparation, the instruction templates are fed back into the LLM to iteratively refine the dataset, avoiding redundancy and improving the dataset's diversity over time.

Labeling of Ground-Truth Data. After obtaining the instruction templates, we randomly generate the target object from detection class synonyms, speed values, object bounding box, etc., and substitute them into the templates. Then, the robot states are determined based on the values in the templates ruled by the robot function logic, and the ground-truth V_m is obtained according to Table I to produce the final training data. This structured generation pipeline broadens the coverage of motion and observation patterns that may occur during real-world deployment.

Interface-Level Sim-to-Real Transfer. LOVON transfers the structured perception–action interface rather than raw simulator pixels or simulator-specific robot dynamics. In both simulation and real-world deployment, visual observations are converted into the same input representation comprising the target class, detection confidence, normalized bounding-box center and size, mission instruction, and task states. L2MM maps this representation to the same physical velocity command $V_m = [v_x, v_y, \theta]$. On real robots, V_m is executed by each platform's native low-level controller, which handles platform-specific kinematics and dynamics. This abstraction reduces dependence on simulator appearance and a particular robot model; the remaining perception discrepancies, including detection noise and locomotion-induced blur, are handled by the Laplacian filtering and target re-search logic and are assessed in the multi-platform real-world experiments.

The dataset preparation process is fast and easy to expand. It takes less than 15 minutes to process 1 million samples using an Intel i9-12900KF CPU.

VII. Experiment

A. Experiment Setup

Model Details. For object detection, L2MM is designed to take inputs in a general string format, which can be directly obtained from normal detection models. Thus, LOVON is independent of detection models. We employ the recently developed detection model, YOLO-World [16], which offers open-vocabulary detections with high performance and lightweight architecture. As the task planner and data generation assistant, we utilize DeepSeek R1 [28]. L2MM is a transformer-based model with feature dimension 256, 4 layers, and 8 attention heads, a feedforward dimension of 1024, and 1 linear layer

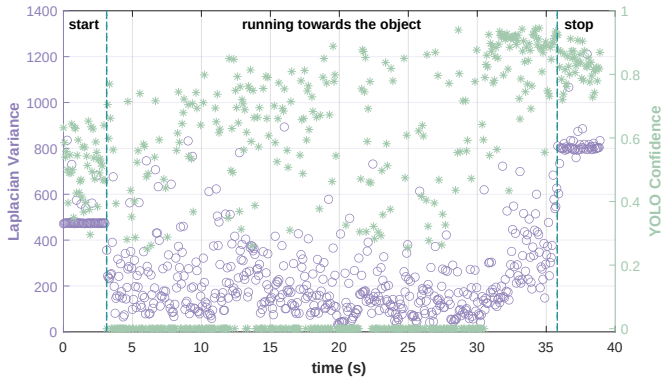


Fig. 6. **Visualization of Laplacian variance and object confidence (0.3 m/s toward chair case).** The robot remains stationary during start and stop phases, and moves toward a target at 0.3/0.5/0.7 m/s during movement. The front camera runs at $\sim$ 15 Hz while keeping the object in view. We compute the Laplacian variance and detection confidence for each captured frame.

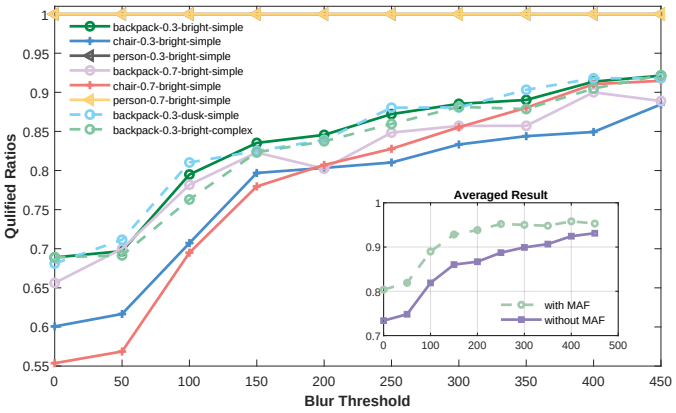


Fig. 7. **Thresholds and qualified ratios of different objects at different speeds, lights, and backgrounds.** Increasing the blur threshold directly leads to a higher qualified frame ratio, but setting the threshold too high can lead to unnecessary rejection of valid frames. Our filter method with MAF increases the qualified frame rate by 21% with $T_{blur} = 150$.

as decoder for each head. IOE shares the same general architecture, with a reduced feature dimension size of 64, 2 transformer layers with 4 attention heads, and a feedforward dimension of 256.

Training Settings. Our dataset consists of 1 million samples, which are divided into training and testing sets in a 4:1 ratio. We use the NVIDIA RTX 3080 Ti GPU for training. The L2MM model is trained with a dropout rate of 0.1, a learning rate of 10^{-4}, a batch size of 512, a max-sequence length of 64, and a motion loss coefficient β set to 10. It is trained for 25 epochs using the AdamW optimizer. The total training time is approximately 1 hour. Similarly, the IOE model is trained with the same settings in approximately 30 minutes.

Robot Settings. LOVON is versatile and can be applied to various legged robots. In our experiments, we evaluate three representative models: Unitree Go2, B2, and H1-2, as shown in Fig. 1-(c). The L2MM velocities V_m are directly fed to Unitree's API for low-level velocity tracking. For the computing platform, we utilize the Jetson Orin, while the visual platform consists of the robots' built-in cameras and the RealSense D435i camera. The IOE achieves 100% accuracy on the training and test sets for object class extraction in $\sim$ 5 ms, compared with $>$ 1000 ms for an LLM API. The inference time of the LLM task planner is about 1 s. The detection model and L2MM update onboard at a frequency of about 5 Hz.

B. Performance of Motion-Blurred Frame Filtering

To investigate the effect of motion blur on object detection, we analyze the correlation between Laplacian variance and detection confidence. The robot approaches targets (backpack, chair, person) at 0.3/0.5/0.7 m/s with a camera frame rate of $\sim$15 Hz, while keeping targets in the field of view. We calculate Laplacian variance and detection confidence for each captured frame. As shown in Fig. 6, both values fluctuate significantly during movement, and low detection confidence strongly coincides with low Laplacian variance. Even with targets in view, the detector frequently yields zero confidence due to motion blur, which degrades image quality and impairs detection performance.

To address motion blur, we apply our Laplacian variance-based frame filtering method. As validated in Fig. 7 across diverse objects, speeds, illuminations, and backgrounds, a higher blur threshold generally increases the qualified frame ratio, but an overly high threshold risks discarding valid sharp frames and reduces filtering effectiveness. The controlled threshold sweep identifies $T_{\text{blur}}^{(0)} = 150$ as an observation-based initialization, which improves the qualified frame ratio by approximately 10% under all tested conditions; during online operation, this value is updated every 2 s using the adaptive rule in Sec. IV-C. We further integrate this filter into the detection pipeline by replacing blurred frames with the latest valid frame and smoothing confidence with a window-3 confidence moving average filter (MAF). The controlled $T_{\text{blur}} = 150$ configuration lifts the overall qualified frame rate by 21%, demonstrating the effectiveness of the filtering pipeline while retaining a reproducible reference setting.

C. Evaluation in Simulation Environments

Benchmark and Evaluation Metric. In our evaluation, we follow the setup from previous works [30] under the Gym-Unreal benchmark [6] (as shown in Fig. 8), where the maximum episode length is 500 steps. The visible region of the tracker is defined as a 90-degree fan-shaped sector with a radius of 750 cm. We evaluate performance using two metrics: 1) Episode Length (EL), which measures the average duration of episodes over 100 trials, reflecting the tracker's long-term performance; 2) Success Rate (SR), which calculates the percentage of successful episodes across 100 trials, indicating the model's robustness.

Baselines. 1) DiMP uses a pre-trained video tracker for bounding box and tunes an optimization-based controller to control the tracker. **2) SARL** trains an end-to-end Conv-LSTM network via online reinforcement learning (RL). **3) AD-VAT+** applies a two-stage training strategy to learn in complex environments with obstacles. **4) TS** uses a pose-based teacher to provide online suggestions to the vision-based student while interacting with environments. **5) RSPT** utilizes RGB-D and combines a video tracker to construct a

TABLE II
QUANTITATIVE RESULTS COMPARED WITH BASELINES IN GYM-UNREAL. THE VALUES IN EACH CELL REPRESENT THE AVERAGE EL AND SR. TRACKVLA AND TRACKVLA++ ARE THE LATEST LEGGED-SPECIFIC VISION-LANGUAGE NAVIGATION METHODS. **MEAN-1**: MEAN VALUE ACROSS DIFFERENT SCENES FOR PERSON TRACKING. **MEAN-2**: MEAN VALUE ACROSS DIFFERENT ANIMAL TRACKING.

Methods	Training time	ParkingLot	UrbanCity	UrbanRoad	SnowVillage	Mean-1	Horse	Dog	Sheep	Mean-2
DiMP [31]	0 hours*	327 / 0.48	401 / 0.66	308 / 0.33	301 / 0.43	334.25 / 0.48	-/-	-/-	-/-	-/-
SARL [32]	24 hours	301 / 0.22	471 / 0.86	378 / 0.48	318 / 0.31	367.00 / 0.47	-/-	-/-	-/-	-/-
AD-VAT+ [33]	24 hours	439 / 0.60	497 / 0.94	471 / 0.80	365 / 0.44	443.00 / 0.70	-/-	-/-	-/-	-/-
TS [34]	26 hours	472 / 0.89	496 / 0.94	480 / 0.84	424 / 0.63	468.00 / 0.83	-/-	-/-	-/-	-/-
RSPT [35]	12 hours	480 / 0.80	**500 / 1.00**	**500 / 1.00**	410 / 0.80	472.50 / 0.90	-/-	-/-	-/-	-/-
EVT [30]	1 hour	484 / 0.92	**500 / 1.00**	496 / 0.96	471 / 0.87	487.75 / 0.94	**500 / 1.00**	469 / 0.90	471 / 0.93	480 / 0.94
TrackVLA [21]	360 GPU-hours	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500.00 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**
TrackVLA++ [24]	192 GPU-hours	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500.00 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**
LOVON (Ours)	**1.5 GPU-hours**	**500 / 1.00**	**500 / 1.00**	499 / 0.99	**500 / 1.00**	499.75 / **1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**	**500 / 1.00**

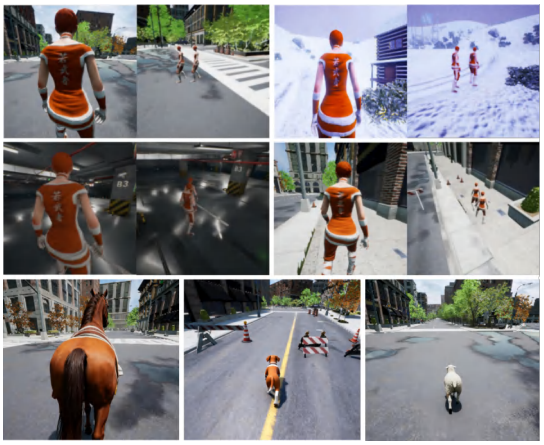


Fig. 8. **Simulation evaluation.** We conduct extensive experiments in Gym-Unreal to comprehensively test the model's robustness and generalization. The evaluation covers two key dimensions: 4 distinct scenes (UrbanCity, SnowVillage, ParkingLot, UrbanRoad) and 3 additional target objects (Horse, Dog, Sheep) to assess performance beyond common benchmark objects.

TABLE III
MODEL SIZE.

Model Name	Size
DiMP [31]	DiMP50 (364 MB)
AD-VAT+ [33]	AD-VAT-main (3.2 MB)
TS [34]	TS-VAT-main (3.2 MB)
RSPT	RSPT-main (3.1 MB)
EVT [30]	DEVA (277 MB), Grounding DINO (694 MB), EVT-main (2.7 MB)
TrackVLA [21]	EVA-CLIP + Vicuna-7B + Heads (13.5 GB+)
TrackVLA++ [24]	SigLIP + DINOv2 + Qwen2-7B + TIM/Heads (14.5 GB+)
LOVON (Ours)	YOLO-World (141 MB), IOE (591 KB), L2MM (13.3 MB)

structure-aware representation and train the policy via RL. **6) EVT** trains a pre-trained visual foundation model with offline RL. **7) TrackVLA** combines an LLM recognition head with diffusion-based trajectory planning and is deployed on a Unitree Go2 quadruped. **8) TrackVLA++** augments a VLA model with spatial reasoning and target-identification memory.

Performance Comparisons. As shown in Table II, our LOVON outperforms several baseline approaches, achieving a perfect SR of 1.00 across most environments, including ParkingLot, UrbanCity, SnowVillage, Horse, Dog and Sheep series. Compared with the strong EVT baseline [30], LOVON demonstrates superior tracking performance, e.g., 500 / 1.00 vs. 484 / 0.92 in ParkingLot, as a result of its integrated searching capability. Moreover, LOVON outperforms EVT in all animal tracking situations. Fig. 9 shows successful and failure

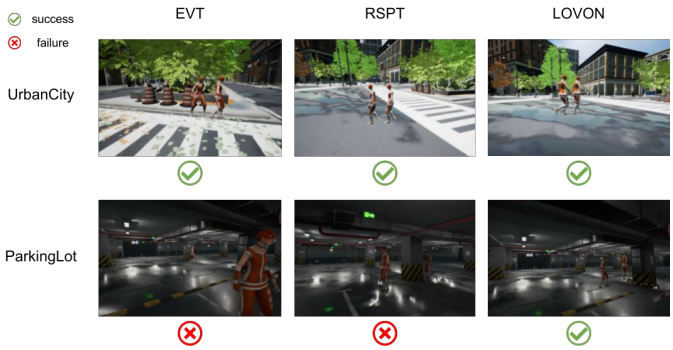


Fig. 9. **Tracking failure cases.** The baseline models (e.g., EVT, RSPT) perform better in the UrbanCity scene than in other scenes. Failure cases in the ParkingLot scene are mainly due to losing the target person. LOVON's integrated searching capability boosts its tracking performance in cluttered scenes, where alternatives often fail to recover tracking after target loss.

cases of the tracking process in simulation. All the baseline models (e.g., EVT, RSPT) perform well in the UrbanCity scene because the space is large and the target person is not easy to lose. However, most of the baseline models do not perform well in scenes like ParkingLot (Table II). These models often fail when the target person turns rapidly and the tracker loses sight of the target. LOVON can search the environment after target loss and successfully resume tracking. Compared with the legged VLA baselines TrackVLA [21] and TrackVLA++ [24], LOVON achieves comparable, near-saturated tracking performance in Gym-UnrealCV with substantially lower training cost. LOVON reaches 499.75/1.00 for Mean-1 and 500/1.00 for Mean-2, closely matching the 500/1.00 aggregate results of both baselines, while requiring 1.5 GPU-hours on a single RTX 3080 Ti, compared with 360 and 192 H100 GPU-hours for TrackVLA and TrackVLA++, respectively. Table III further shows that LOVON uses compact task-specific modules (13.9 MB for IOE and L2MM in total), whereas both VLA baselines rely on 7B-scale language backbones. These results demonstrate LOVON's favorable accuracy-efficiency trade-off.

D. Evaluation on Real-World Experiments

We perform quantitative real-world evaluations on three legged robots with distinct mechanical designs (Unitree Go2, B2, and H1-2) using human-targeted navigation tasks. All experiments rely on native robot controllers, with $T_{\text{blur}} = 150$,

TABLE IV
REAL-WORLD EVALUATIONS OF THREE ROBOT TYPES UNDER DIVERSE TEXTURED SCENARIOS, ILLUMINATION CONDITIONS, TARGET TASKS AND COMMAND SPEEDS. CS: COMMAND SPEED; N_s: NUMBER OF SEARCHING TIMES; T_s: TASK TIME; SR: SUCCESS RATE, AVERAGE OF 20 TESTS.

Robot	Environment	Lightness	Target	Distance	Speed (m/s)	T_{blur}	N_s	T_s (s)	SR (%)
Go2	Lab	Indoor Bright	Person	4 m	0.30	150	1.00	17.47	100
B2	Lab	Indoor Bright	Person	4 m	0.30	150	1.00	18.03	100
H1-2	Lab	Indoor Bright	Person	4 m	0.30	150	1.00	18.19	100
Go2	Sandlot	Outdoor Bright	Ball	$\leq$ 5 m	0.30	150	1.15	-	100
Go2	Lab	Indoor Bright	Umbrella	$\leq$ 6 m	0.38	150	1.05	-	100
Go2	Parking Lot	Outdoor Bright	Car	$\leq$ 10 m	0.70	150	1.00	-	100
Go2	Parking Lot	Indoor Low-light	Car	$\leq$ 8 m	0.55	150	1.10	-	100
Go2	Road	Outdoor Dusky	Person	$\leq$ 8 m	0.65	150	1.00	-	100
Go2	Hallway	Outdoor Dusky	Potted Plant	$\leq$ 6 m	0.45	150	1.10	-	100
Go2	Grass	Outdoor Bright	Moving Person	$\leq$ 6 m	0.80	150	1.25	-	100
Go2	Stairs	Indoor Bright	Moving Person	$\leq$ 3 m	0.80	150	1.40	-	90
Go2	Lab	Indoor Bright	Backpack,Chair,Person	$\leq$ 6 m	0.38	150	1.35	-	95
Go2	Lab	Indoor Bright	Backpack,Chair,Dustbin,Person,Umbrella	$\leq$ 4 m	0.30	150	1.65	-	90

a commanded speed of 0.3 m/s, and a target distance of 4 meters in the laboratory environment (Fig. 1). Each configuration is tested 20 times, and average results are reported in Table IV. All robots achieve $N_s = 1.00$ and 100% success rate (SR), verifying the consistent and robust performance of LOVON across different hardware platforms. Go2 completes the task in the shortest time ($T_s = 17.47$s) due to its compact structure and superior motion control, while B2 (18.03s) and H1-2 (18.19s) exhibit slightly longer execution times.

To comprehensively validate the versatility and robustness of LOVON, extensive real-world experiments are conducted across diverse scenarios, as summarized in Table IV. The evaluations cover **multiple robot platforms**, **varied environments** (laboratory, sandlot, parking lot, road, hallway, grass, and stairs), **different illumination conditions** (indoor bright, outdoor bright, indoor low-light, outdoor dusky), **diverse target objects** (person, ball, umbrella, car, potted plant, backpack, chair, and dustbin), **various command speeds** (0.30–0.80 m/s), **complex backgrounds** with different textures, and **sequential-subgoal tasks**. Real-world implementation also confirmed interface compatibility: YOLO-World and Grounding DINO were interchangeable after conversion to the common detection tuple, while DeepSeek-R1, doubao-1-5-lite-32k-250115, and Qwen2.5-7B-Instruct all produced correctly ordered subtasks using the same fixed prompt. These checks verify the replaceability of the upstream backbones rather than comparative performance among LOVON components.

(1) Cross-platform Deployment. Tests on Unitree Go2, B2, and H1-2 (Fig. 1-(c)) validate the platform-agnostic nature of LOVON, achieving 100% success rate across all three systems.

(2) Open-world Object Adaptation. LOVON generalizes to everyday objects of varying sizes (Fig. 1-(e)), including large targets (cars), medium targets (humans), and small objects (backpacks, umbrellas), enabling stable interaction in unstructured environments.

(3) Long-horizon Multi-goal Tracking. Enabled by the LLM planner, LOVON realizes long-distance object navigation (Fig. 1-(a)) and supports sequential multi-target tracking, maintaining robust performance in complex environments.

(4) Dynamic Target Tracking. LOVON performs reliable dynamic tracking (Fig. 1-(b)) on moving targets, tested on flat ground, spiral stairs, and grassland. Most tasks achieve 100% success rate, with the stair scenario reaching 90%.

(5) Disturbance Robustness. LOVON maintains resilience to occlusions, target displacement, and physical perturbations (Fig. 1-(d)). The robot can re-localize targets after interference and complete tasks in challenging terrain such as sandy playgrounds, where mechanical jolting is severe.

Quantitative Failure and Re-localization Analysis. We quantify failure rates, target loss frequency, and recovery times based on Table IV. The stair dynamic tracking task (90% SR) fails mainly because the target moves rapidly on stairs, causing temporary loss under jolting. The three- and five-subgoal tasks achieve 95% and 90% SR, respectively. Their failures mainly arise when accumulated positioning and heading errors across successive subgoals alter the camera viewpoint and cause the next target to be occluded. The metric N_s directly quantifies target loss frequency under mechanical disturbances (e.g., $N_s = 1.15$ on sand). Transient losses are automatically corrected by the model. For prolonged target loss, the robot typically rotates 360° to re-localize, taking approximately 25s per search cycle. Our Laplacian filtering effectively suppresses motion blur from high-frequency shaking, ensuring stable tracking and fast recovery even under severe terrain disturbances.

E. Ablation Study

Ablation on the Model Parameters. We investigate the impact of model size, dataset size N_{ds}, motion loss weight β, and the inclusion of special tokens. The model's performance is evaluated by comparing the standard deviation σ_v and the average speed bias ϵ_v compared to the speed given in the instruction (0.40 m/s). As shown in Table V, the base-size model shows good performance, while smaller models exhibit higher σ_v and ϵ_v, indicating they cannot effectively capture the required information. Larger models generally perform better, but the improvement in speed tracking is marginal despite the model size being much larger. In addition, models trained on smaller datasets show lower ϵ_v but higher σ_v, reflecting instability. Larger datasets improve performance but do not significantly impact real-world applicability. The motion loss weight β plays a crucial role. A smaller $\beta = 1$ results in poor performance, as the model gives insufficient

TABLE V
ABLATION STUDY ON MODEL PARAMETERS. WE TRAIN ALL THE MODELS FOR 50 EPOCHS AND SELECT THE MODEL WITH THE LOWEST VALIDATION LOSS FOR EVALUATION.

Model Size	N_{ds} (m)	β	[SEP]	$\sigma_v(\times 10^{-4})\downarrow$	$\epsilon_v(\times 10^{-2})\downarrow$
LOVON-Small (0.47M)	0.4	10	✓	3.74	5.74
LOVON-Large (25.51M)	0.4	10	✓	0.43	0.23
LOVON-Base (3.30M)	0.2	10	✓	3.73	0.43
	0.8	10	✓	0.40	1.49
	0.4	1	✓	4.59	7.88
	0.4	20	✓	Fail	Fail
	0.4	10	×	1.02	29.5
LOVON-Base (Best)	**0.4**	**10**	✓	**0.38**	**1.80**

TABLE VI
ABLATION STUDY ON PROPOSED METHODS (20 AVERAGE). N_s: NUMBER OF SEARCHING; T_s: SEARCH TIME (S)

Object	States	Filter	4 m		6 m	
			$N_s\downarrow$	$T_s\downarrow$	$N_s\downarrow$	$T_s\downarrow$
Backpack	3	×	4.05	100.00	5.95	178.00
	4	×	1.60	45.91	2.25	83.09
	4	✓	1.00	22.25	1.00	32.57
Chair	3	×	2.05	43.50	4.25	110.00
	4	×	1.05	26.29	1.35	45.34
	4	✓	1.00	21.73	1.00	30.55
Person	3	×	1.00	19.80	1.00	28.00
	4	×	1.00	17.66	1.00	26.24
	4	✓	1.00	17.47	1.00	26.51

attention to motion loss. Conversely, a larger $\beta = 20$ leads to undervaluation of state loss, causing inaccurate state inference. Finally, the inclusion of the special token [SEP] is essential. The model trained without this token struggles to follow the given speed, as [SEP] helps differentiate between various input components, especially language.

Ablation on the Filter Method and the Number of States. We evaluate the impact of the number of searching states and the frame-filtering technique on the efficiency of object navigation when the target is lost. Three system configurations are tested: Case 1, with three states and no frame filtering; Case 2, with four states but no frame filtering; and Case 3, with four states and frame filtering. Experiments are conducted on the Go2 robot with three objects (backpack, chair, and person) at distances of 4m and 6m, setting the speed in the instruction to 0.3 m/s. As shown in Table VI, LOVON excels in seeking the person, which is easily detected, while the backpack, which is harder to track, requires more effort. In Case 1, the robot performs inefficiently, often losing the object and requiring significant time due to motion blur. Adding a mission state in Case 2 improves efficiency by enabling the robot to search in both directions, though object loss and shaking still occur, leading to a higher number of searching circles. In Case 3, with four states and the frame-filtering technique, LOVON achieves optimal efficiency. It reduces N_s to 1 for both the backpack and chair, matching the performance with the person. Furthermore, the time to reach the target is reduced by 5 and 2 times for the backpack and chair, respectively, compared to Case 1, approaching the performance observed for the person.

VIII. CONCLUSION

In conclusion, we introduce LOVON, a model that bridges long-horizon open-vocabulary vision-language navigation with practical legged robot execution. Taking vision and language perceptions as inputs, LOVON is able to plan long-horizon tasks and output language-specified velocity for direct compatibility by leveraging LLM and L2MM. To address motion-induced perception degradation, we incorporate the Laplacian Variance Filter and a window-3 MAF, which significantly enhance the model's performance in real-world applications. Quantitative evaluations across three different legged robots and diverse scenarios demonstrate LOVON's cross-platform adaptability. Extensive simulations and real-world experiments across a wide range of complex environments demonstrate the robustness and effectiveness of LOVON.

Limitations and Future Work. Although the online blur-threshold update and window-3 MAF improve adaptability, they remain lightweight rules based on recent visual observations and may be affected by abrupt illumination changes, texture-poor scenes, or persistent blur; future work will jointly use visual statistics and robot-motion cues for stronger online adaptation. In addition, because LOVON does not maintain an explicit global map, positioning and heading errors may accumulate over longer subgoal sequences and increase target research; lightweight spatial memory and active re-localization will be explored to improve long-horizon reliability.

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